Unit 11, Lesson 3: Evaluating Algebraic Expressions Using Substitution

Benchmark

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

Learning Target

☐ I can evaluate algebraic expressions using substitution.

11.3.1 Warm-Up: Order Explorer

1. Place a digit in each box to create an expression with the largest possible value. Use any of the digits 1 to 5, once each at most.

$$\square^{\square} + \square \times \square$$

2. What is the value of your expression?

3. On a scale of 1 to 10, how confident are you that your expression is the largest possible value?

4. Explain your strategy for choosing and placing the numbers as you did.

11.3.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers

1. Brae evaluated the expression $7b + 12 - (5 - 3)^2$ for $b = -2$. Her work is shown.

    a. Evaluate her work and determine where she made mistakes.

$$
\begin{gathered}
7b + 12 - (5 - 3)^2 \\
7 - 2 + 12 - (5 - 3)^2 \\
7 - 2 + 12 - (-2)^2 \\
7 - 2 + 12 + 4 \\
5 + 12 + 4 \\
17 + 4 \\
21
\end{gathered}
$$

    b. Help Brae understand and learn from her mistakes. Write a brief note to Brae explaining what her mistakes were and how she could learn from the mistakes.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

c. Read the note your partner wrote and offer suggestions on ways to improve their note.

d. Using the word bank below, complete the three reminders that could help Brae.

inside	outside	parentheses

- When substituting values into algebraic expressions, it is helpful to put the substituted value(s) in ________________.
- When evaluating expressions with exponents, if the negative is ____________ the parentheses, the negative indicates subtraction, or the opposite of the value found by raising the value inside to a power.
- If the negative is ________________ the parentheses, it indicates that the number is a negative number being raised to a power.

e. Verify your answers with your partner and, if necessary, resolve any differences.

2. Complete the following problems.

a. Evaluate the expression $-8m - 4q^2 + \frac{s}{3}$ where $m = 4,\ q = -3$, and $s = -24$.

b. Evaluate the expression $3s - 4q - m^2$ where $m = 4,\ q = -3$, and $s = -24$.

c. Evaluate the expression $-m^3 - 3q + (s - q)$ where $m = 4,\ q = -3$, and $s = -24$.

d. All of the expressions should have the same result. Check your work with your partner.

11.3.3 Your Turn!

1. What is the value of $5(6x + 7) - 3y^2$ if $x = -1$ and $y = 5$?

2. Evaluate the expression $3d^4 - \frac{W}{3}$ where $d = 4$ and $W = -75$.

3. Find the value of $-2(b + 4) + 3a - (7 - a)^5$ if $a = -3$ and $b = 5$.

4. Evaluate the expression $c^3 - 5k + (h - k)$ where $c = -9,\ h = 29$, and $k = -17$.

5. Evaluate each expression.

   a. $8 \div p - 16 \div (-5 + 7)$ where $p = 4$

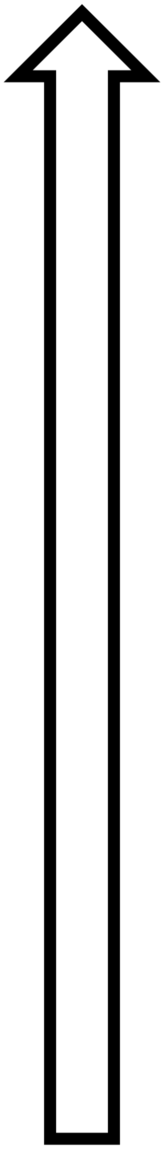
   b. $13 + \frac{6}{y} - y^3$ if $y = -3$

   c. $4g - 12 + (h - 7) \cdot 3^2$ where $g = -7$ and $h = 6$

   d. $w \div t \cdot H \div y^2$ where $H = -3, t = 7, w = 84$, and $y = -2$

11.3.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding.



I'm so confident, I can explain evaluating algebraic expressions using substitution to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute a value

I tried hard, but I am finding this challenging. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute in a value